Efficient Rationale-based Retrieval: On-policy Distillation from Generative Rerankers based on JEPA

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Abstract

Unlike traditional fact-based retrieval, rationale-based retrieval typically necessitates cross-encoding of query-document pairs using large language models, incurring substantial computational costs. To address this limitation, we propose Rabtriever, which independently encodes queries and documents, while providing comparable cross query-document comprehension capabilities to rerankers. We start from training a LLM-based generative reranker, which puts the document prior to the query and prompts the LLM to generate the relevance score by log probabilities. We then employ it as the teacher of an on-policy distillation framework, with Rabtriever as the student to reconstruct the teacher's contextual-aware query embedding. The Joint-Embedding Predictive Architecture (JEPA) paradigm is adopted, which integrates a lightweight, trainable predictor between LLM layers and heads, projecting the query embedding into a new hidden space, with the document embedding as the latent vector. JEPA then minimizes the distribution difference between this projected embedding and the teacher embedding, with an auxiliary loss on the reverse KL of LLM logits, to reshape the student's logit distribution. Rabtriever optimizes the teacher's quadratic complexity on the document length to linear, verified both theoretically and empirically. Experiments show that Rabtriever outperforms different retriever baselines across diverse rationale-based tasks, including empathetic conversations and robotic manipulations, with minor accuracy degradation from the reranker. Rabtriever also generalizes well on traditional retrieval benchmarks such as MS MARCO and BEIR, with comparable performance to the reranker, as well as the best retriever baseline.

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CCS Concepts

• **Information systems** → **Language models**; • **Computing methodologies** → *Learning from demonstrations*; *Information extraction*; Learning latent representations.

Keywords

Rationale-based Retrieval, On-policy Distillation, JEPA

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1 Introduction

Dense retrieval has emerged as a dominant paradigm of information retrieval (IR), which encodes both queries and documents as dense embeddings within a shared latent space, with relevance typically measured via semantic textual similarity [1, 19, 23]. Nevertheless, the growing adoption of retrieval-augmented generation (RAG) systems increasingly demands more sophisticated forms of knowledge comprehension, requiring retrievers to move beyond the mere matching of explicit facts toward nuanced, often implicit, interpretations of knowledge [3, 4, 26, 27]. This evolution aligns with the knowledge stratification framework proposed by [27], which defines four ascending levels of retrieval complexity and depth:
• *Level-1: Explicit Facts*, • *Level-2: Implicit Facts*
• *Level-3: Interpretable Rationales*, • *Level-4: Hidden Rationales*
While L1 and L2 correspond to conventional **fact-based retrieval**, L3 and L4 call for **rationale-based retrieval**, where the system must comprehend the underlying reasoning, decision logic, or expert knowledge that connects a query to a relevant document. For instance, consider *a software developer who derives actionable coding decisions not only from API documentations alone, but also by examining detailed debugging histories.* In such advanced retrieval scenarios,

the relationship between query and document often extends far beyond straightforward semantic or paraphrastic similarity, exposing a fundamental limitation of mainstream dense retrievers [8].

Leveraging the extensive domain knowledge and sophisticated reasoning capabilities, LLMs have recently been adopted as backbone architectures for dense retrievers, offering a promising avenue for rationale-based retrieval. LLM-based **retrievers** typically approach this challenge either through prompting strategies [10] or integrating with generative objectives [15, 25]. Nevertheless, they still encode queries and documents independently during inference, lacking a contextualized understanding of their online interactions. In contrast, LLM-based **rerankers** [8, 12, 14, 21] jointly encode the concatenated query–document input, thereby enabling deeper and fine-grained reasoning about rationale-based relevance. However, such rerankers require online encoding of every query–document pair, incurring prohibitive computational overheads, which limit their feasibility for dense retrievers in large-scale retrieval systems. Instead, rerankers are generally deployed as the downstream of retrievers in multi-stage text retrieval [12].

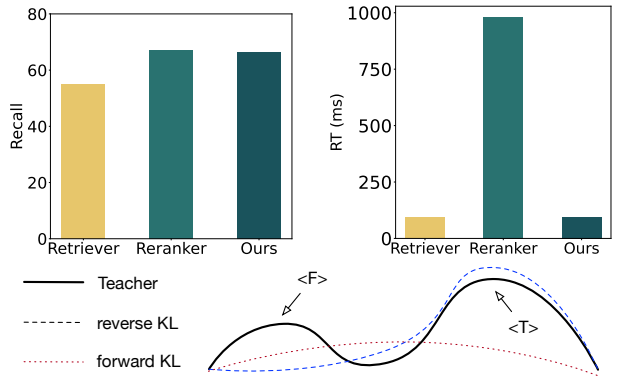


Figure 1: Top: By effective distillation, Rabtriever achieves a similar accuracy to Reranker and a similar speed to Retriever. Bottom: The teacher has two peaks (<T> and <F>) after fine-tuning; forward KL tends to learn averaged behaviors, while reverse KL helps the student seek the most typical mode [5].

To resolve this tension, the purpose of this study is to provide a reliable and practical rationale-based retriever, **with comparable retrieval accuracies to rerankers, and similar inference speeds to traditional retrievers**. The core of our approach lies in an on-policy distillation[1] strategy, which transfers cross-comprehension capability of a reranker **teacher** into a retriever **student**, reconstructing the reranker's embedding and logit distributions by lightweight cross-computations. To achieve this purpose, we adopt the relevance generation paradigm proposed by Sun et al. [21], which puts the document *prior to* the query within the textual input, and produces the relevance score by a generative task. The casual attention mechanism of LLM then naturally ensures the document embeddings are identical between **teacher** and **student**, such that they can be pre-computed offline.

In this paper, we first finetune a **teacher** based on the LaHoRe framework [8], which prompts the LLM to answer "True" or "False" via a binary-choice instruction, and yields the relevance score as the relative log-probability ratio on binary answers. After that, we implement the **student** with the same LLM backbone, with parameter initiated from the **teacher** and set frozen. We then leverage a self-supervised framework called Joint-Embedding Predictive Architecture (JEPA) [2], in which a trainable **predictor** is inserted between LLM layers and heads, to project the **student**'s query embedding into a latent space that is implicitly conditioned on the document embedding. We implement the predictor with a 2-layer MLP and an elementary-multiplication operator to ensure it is sufficiently lightweight. Finally, we adopt both feature-based and logit-based paradigms of white-box knowledge distillation [13], where the first corresponds to the energy-based modeling (EBM) framework of JEPA, by minimizing MSE between **teacher** and **student** embeddings; and the second is an auxiliary loss term to reduce the reverse KL between **teacher** and **student** logits, which ensures the student focus on target tokens (*i.e.*, "True" or "False") instead of spreading across suboptimal vocabularies [5] (Figure 1 Bottom). On the inference stage, we reconstruct the distribution of the **teacher**'s context-aware query embedding by inferring the predictor, and compute the final relevance score. Figure 2 visualizes this paradigm, and we name it by **Ra**tionale-**B**ased Re**triever**, or **Rabtriever** for abbreviation.

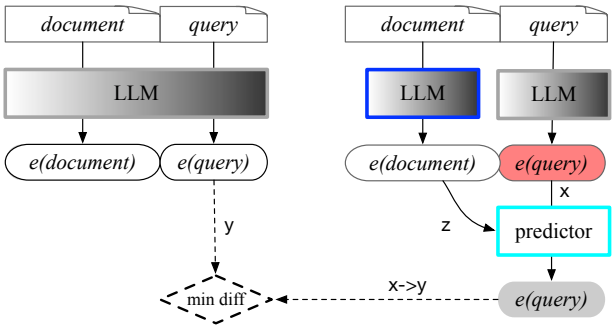


Figure 2: Block diagram visualization of our paradigm. The reranker (left) jointly encodes document and query, with all computations online; Rabtriever (right) encodes document and query separately, with the query embedding unconditioned on document (marked in red). We adopt the JEPA paradigm, where a predictor first converts the query embedding (e_x) into a new latent space ($e_{x\to y}$) with the document embedding played as the latent control (z), then the difference energy between $e_{x\to y}$ and the teacher's query embedding (e_y) is minimized. The blue outline indicates the offline computation; the cyan outline indicates lightweight calculation.

Following Ji et al. [8], we first conduct our experiments on two challenging rationale-based retrieval scenarios:
(1) A conversational agent identifying the optimal dialogue strategy from an annotated pool to enhance interaction quality, based on specific conversational goals (*e.g.*, emotional support) [11, 20].
(2) A language-grounded robot selecting the most appropriate skill to accomplish a high-level instruction, requiring an understanding of procedural intent and environmental constraints [6, 7, 18].

[1] https://thinkingmachines.ai/blog/on-policy-distillation/

The results substantiate that Rabtriever matches the retrieval accuracy of state-of-the-art (SoTA) rerankers while achieving orders-of-magnitude reduction in inference latency, as highlighted by Figure 1 (Top). To further verify Rabtriever's generalization across standardized large-scale retrieval benchmarks, we then evaluate it by both in-domain retrieval on MS MARCO passage ranking, as well as zero-shot retrieval on the BEIR benchmark. Rabtriever performs similarly to the most powerful retriever competitor, indicating that our framework generalizes to traditional retrieval tasks. In summary, the principal contributions of this work are fourfold:
(1) We propose an efficient LLM-based retriever, which encodes queries and documents independently to ensure efficiency, while performing similarly to rerankers on rationale-based retrieval.
(2) We design a distill paradigm on a generative reranker teacher, and employ JEPA to reconstruct its document-conditioned query embedding, by minimizing the latent distribution discrepancy.
(3) We conduct comprehensive evaluations across diverse scenarios, including rationale-based tasks (empathetic conversation and language-grounded robotics) and two general retrieval benchmarks.

2 Related Works

LLM-based dense retrievers. Prevail LLM-based retrievers employ the LLM layers to encode query and document separately, and calculate the relevance score by embedding similarity. For example, RepLLama [12] gets the last-token embedding of Llama and finetunes with the infoNCE loss. E5 [23] further improves the performance by augmenting the training sets with diverse, multilingual synthetic datasets. Promptriever [24] adds the prompt mechanism on RepLLama with performance improvements on instructed retrieval tasks. Recently, hybrid retrievers have been proposed, with a unified architecture on both representative and generative tasks [15, 25]. However, these retrievers are primarily designed for fact-based retrieval, which significantly differs from rationale-based retrieval, the main focus of this paper. Although Llama2Vec [10] tries to bridge the gap by incorporating a simple prompt ("NEXT: ") into the query encoding, explicit comprehension on the cross semantics between query and document is still missing.

LLM-based rerankers. Within the paradigm of multi-stage text retrievals, the reranker provides a more accurate ranking by jointly encoding the query-document pair, performing the downstream of the retriever. Its ranking score might be obtained either from the log probability, such as SGPT [14] and rankGPT [21], or by a linear projection like rankLlama [12]. In this paper, we implement a generative reranker similar to LaHoRe [8], which uses a relevance generation paradigm similar to [21], while conducting the finetuning as in [12]. With SoTA performances obtained on several rationale-based retrieval benchmarks, this reranker is then employed as the **teacher** for further distillation to the efficient retriever, so that the inference speed can be optimized dramatically.

3 Preliminaries

3.1 Joint-Embedding Predictive Architecture

Given an encoded input-output pair (x, y), a joint-embedding predictive architecture (JEPA) [2] first converts x into the latent space of y by a predictor $\mathcal{P}(\cdot) : \mathbb{R}^x \rightarrow \mathbb{R}^y$, then minimizes its distributional discrepancy with y:

$$\min \mathcal{F}(\mathcal{P}_z(x), y) \tag{1}$$

where $\mathcal{F}(\cdot, \cdot)$ is nonnegative and can be viewed as the energy between x and y, *i.e.*, $\mathcal{F}(x, y) \rightarrow 0$ when $x \rightarrow y$. Subsequently, the JEPA framework belongs to the energy-based model (EBM). $\mathcal{P}$ can be controlled by a prior, non-trainable latent vector z.

3.2 Problem Formulation

Given a pool of n document candidates $\{D\}$ and a user query Q, the retriever is required to calculate document relevant scores, $\{s_i(Q, D_i)\}, 1 \leq i \leq n$, then returns the Top-$k$ $\{D_i\}$ with respect to the ranking of s_i ($k \leq n$). According to the annotations in the training set, the ground-truth s_i can be labeled by either positive (1) or negative (0). To simplify the computational analysis, we assume the average length of Q and D can be represented by L_Q and L_D.

In the following subsections, we will revisit the paradigms of typical retrievers and rerankers, with corresponding time complexity analysis. We then propose the formulation of Rabtriever, with a probabilistic view of the training objective. Table 1 summarizes three paradigms. In the table (and also the following formulations), we use blue to denote modules which can be calculated off-the-shelf (offline); and use cyan to denote lightweight calculations.

3.3 Bi-encoder Retriever

Traditional efficient retrievers typically employ a bi-encoder architecture, which encodes D and Q separately:

$$e_D = \mathcal{M}^D(D) \in \mathcal{R}^d, e_Q = \mathcal{M}^Q(Q) \in \mathcal{R}^d, \hat{s}^{\text{be}} = < e_D, e_Q > \tag{2}$$

where $\mathcal{M}^Q$ and $\mathcal{M}^D$ are the query and document encoders, while e_D and e_Q are the resulted query and document embeddings, in the dimension of d; $< \cdot, \cdot >$ is a lightweight similarity function; The hat on top of s indicates the prediction, and 'be' denotes bi-encoder.

Time complexity. Encoding of D can be pre-computed off-the-shelf (offline). Assume the encoder is has a quadratic complexity (*e.g.*, transformer), then the total time complexity of Eq. (2) is

$$\underbrace{n\mathrm{O}(L_D)^2}_{\text{Offline}} + \underbrace{\mathrm{O}(L_Q)^2 + n\mathrm{O}(1)}_{\text{Online}} \tag{3}$$

3.4 Cross-encoder Reranker

Although efficient, the bi-encoder framework lacks the joint comprehension of Q and D, therefore struggles to retrieve a rationale-based query reasonably [8]. Instead, a cross-encoder can jointly study the semantic or logic correlation between Q and D:

$$e^{\text{ce}}_{QD} = \mathcal{M}^{\text{ce}}(Q, D) \in \mathcal{R}^d, \hat{s}^{\text{ce}} = \mathcal{H}^{\text{ce}}(e^{\text{ce}}_{QD}) \tag{4}$$

Time complexity. In practice, $\mathcal{M}^{\text{ce}}$ can also be implemented with transformers (BERT, GPT, etc), with also quadratic time complexity. However, it must conduct all the computations online:

$$n\mathrm{O}(L_D + L_Q)^2 \quad \text{(Online)} \tag{5}$$

Since L_D, L_Q, and n are larger than 1, it is obvious that Eq. (5) is larger than Eq. (3), *i.e.*, the cross-encoder has a much higher time complexity than the bi-encoder.

Table 1: Comparison of different retriever architectures, including Retriever, Reranker, and our Rabtriever. Rabtriever first initializes from a pretrained reranker, then inserts a trainable latent projector $\mathcal{P}$. Instead of predicting the retrieval scores ($\mathcal{L}$) directly, Rabtriever employs the EBM paradigm which aims to reconstruct the latent distribution of the cross-encoding reranker. Modules in blue indicate the offline computation; modules in cyan indicate lightweight calculation.

Definition	query D, document D, embedding e, relevance score s (ground truth) and $\hat{s}$ (predicted); encoder $\mathcal{M}$, and the lightweight calculator $\mathcal{H}$.		
	Bi-encoder **Retriever**	Cross-encoder **Reranker**	**Rabtriever**
Initialize	$\mathcal{M}^{\mathcal{Q}}, \mathcal{M}^{D} \leftarrow$ LLM	$\mathcal{M}^{\text{ce}} \leftarrow$ LLM	$\mathcal{M} \leftarrow \mathcal{M}^{\text{ce}}, \mathcal{H} \leftarrow \mathcal{M}^{\text{ce}}$
Forward	$\hat{s}^{\text{be}} \sim \mathcal{H}^{\text{be}}(s\|e_Q, e_D)\,\mathcal{M}^{\mathcal{Q}}(e_Q\|Q)\,\mathcal{M}^{D}(e_D\|D)$	$\hat{s}^{\text{ce}} \sim \mathcal{H}^{\text{ce}}(s\|e^{\text{ce}}_{QD})\,\mathcal{M}^{\text{ce}}(e^{\text{ce}}_{QD}\|Q, D)$	$\hat{s} \sim \mathcal{H}(\hat{s}\|e_{QD})\,\mathcal{P}(e_{QD}\|e_D, e_Q)\,\mathcal{M}(e_Q\|Q)\,\mathcal{M}(e_D\|D)$
Objective	$\min_{\mathcal{M}^{\mathcal{Q}}, \mathcal{M}^{\mathcal{D}}} \mathcal{L}(\hat{s}^{\text{be}}, s)$	$\min_{\mathcal{M}^{\text{ce}}, \mathcal{H}^{\text{ce}}} \mathcal{L}(\hat{s}^{\text{ce}}, s)$	$\min_{\mathcal{P}} \mathcal{F}(e_{QD}, e^{\text{ce}}_{QD}) + \text{KL}(\hat{s}\|\|\hat{s}^{\text{ce}})$, $\mathcal{F}(x, y) \to 0$ when $x \to y$

3.5 Reconstruct Cross-encoding by Bi-encoder

To tradeoff the pros and cons of bi-encoder (Eq. (2)) and cross-encoder (Eq. (4)), here we propose a new bi-encoder architecture:

$$e_D = \mathcal{M}(D), e_Q = \mathcal{M}(Q), e_{QD} = \mathcal{P}(e_Q, e_D), \hat{s} = \mathcal{H}(e_{QD}) \tag{6}$$

Similar to the nominal bi-encoder, we separately encoded e_Q and e_D to obtain a high computation efficiency. To obtain a comparable retrieval accuracy to the cross-encoder, however, we insert an extra latent predictor $\mathcal{P}$ to project e_{QD} into a new latent space, such that $e_{QD} \to e^{\text{ce}}_{QD}$, *i.e.*, to reconstruct the cross-encoded e^{ce}_{QD} from Eq. (4). Note $\mathcal{M}(D)$ can be computed offline; as long as we design $\mathcal{P}$ and $\mathcal{H}$ to be **lightweight enough**, we obtain a lower time complexity than the cross-encoder. In the next section, we will deduce an energy-based objective which matches the JEPA formulation (Eq. (1)).

4 Method

4.1 Teacher: A Generative LLM-based Reranker

For the **teacher**, we implement the reranker with an LLM backbone, which accepts a textual input and produces next-token probabilities:

$$\text{P}^{\text{teacher}} \sim \mathcal{H}^{\text{ce}}(\mathcal{M}^{\text{ce}}(Q, D)) := \text{LLM}(\mathcal{I}(D, Q)) \tag{7}$$

where $\mathcal{M}^{\text{ce}}$ are the LLM blocks and $\mathcal{H}^{\text{ce}}$ are the LLM language heads. According to the LaHoRe framework [8], we employ an instruction template $\mathcal{I}$ which puts the document before the query:

> **Document**: {D}
> **Query**: {Q}
> Can **Query** be appropriately replied with **Document**?
> If the answer is true, choose <T>; otherwise, choose <F>.

The format of the binary-choice question follows the relevance generation paradigm [21], which encourages LLM to thoroughly consider its preference on the relevance, and outputs either <T>or <F>, with corresponding log probabilities. Based on the causal attention mechanism of LLM, **the D part of encoding is equivalent to the separately encoded D:**

$$\text{LLM}(D, Q)[:L_D] = \text{LLM}(D) \tag{8}$$

where [] indicates the indexed selection on the sequence. This property precludes the need for the D-encoder distillation, with only the reconstruction of Q needed.

Training. Similar to [8], we construct the QA-format training samples by augmenting the positive query-document pairs with all possible negative pairs to leverage the scaling of LLM. The question is constructed by incorporating the query and document into the instruction template $\mathcal{I}$; while the answer is annotated by either <T >(for positive pair) or <F >(for negative pair). Then we conduct a standard supervised fine-tuning (SFT) on Eq. (7), with the cross-entropy loss ($\mathcal{L}$) solely on the answer. The retrieval task is then converted to a pure generative task, and the fine-tuned LLM becomes the **teacher** for the subsequent distillation. Figure 3 (I) shows this training mechanism.

Inference. During the inference stage, the relevance score is deduced from the **teacher**'s relative positive-negative confidence:

$$\hat{s}^{\text{teacher}} = \log \text{P}(< T >)/(\log \text{P}(< T >) + \log \text{P}(< F >)) \tag{9}$$

Time complexity. This retriever adheres to the online time complexity of the typical **cross-encoder**: $n\text{O}(L_D + L_Q)^2$, as indicated in Eq. (5). This high time complexity prevents the application of **teacher** as an efficient retriever, making the distillation necessary.

4.2 On-policy Distillation by JEPA

The Architecture. As a bi-encoder **student**, we implement both D and Q encoders with the same LLM backbone, with parameters initialized from the **teacher**, *i.e.*, *i.e.*, $\mathcal{M} \leftarrow \mathcal{M}^{\text{ce}}, \mathcal{H} \leftarrow \mathcal{H}^{\text{ce}}$ according to Eq. (6) and Eq. (4). Given $e_D = \mathcal{M}(D), e_Q = \mathcal{M}(Q)$[2], we then insert a lightweight predictor block $\mathcal{P}$ between $\mathcal{M}$ and $\mathcal{H}$, which produces a new embedding and generates the final retrieval score:

$$e^{\text{student}}_Q = \mathcal{P}_{z=e_D}(e_Q), \quad \text{P}^{\text{student}} \sim \mathcal{H}(e_D \oplus e^{\text{student}}_Q) \tag{10}$$

where z is the latent variable of $\mathcal{P}$ to impose the grounding impact of D on e_Q, and $\oplus$ means vector concatenation. The relevance score $\hat{s}^{\text{student}}$ is calculated from $\text{P}^{\text{student}}$ the same as Eq. (9).

Connection to JEPA.. By minimizing the energy between **teacher** and **student** embedding, the architecture can be considered as the JEPA paradigm:

$$\min_{\mathcal{P}} \mathcal{F}(e_D \oplus \mathcal{P}_{z=e_D}(e_Q), e^{\text{ce}}_{QD}) \tag{11}$$

Compared to the original formulation of JEPA, *i.e.*, Eq. (1), e_Q plays as the input x, e^{ce}_{QD} plays as the output y, and e_D is the latent z. Figure 2 visualizes this distillation paradigm.

[2]Note e_Q is equivalent with LLM(D) in Eq. (8) due to the causal mechanism of LLM.

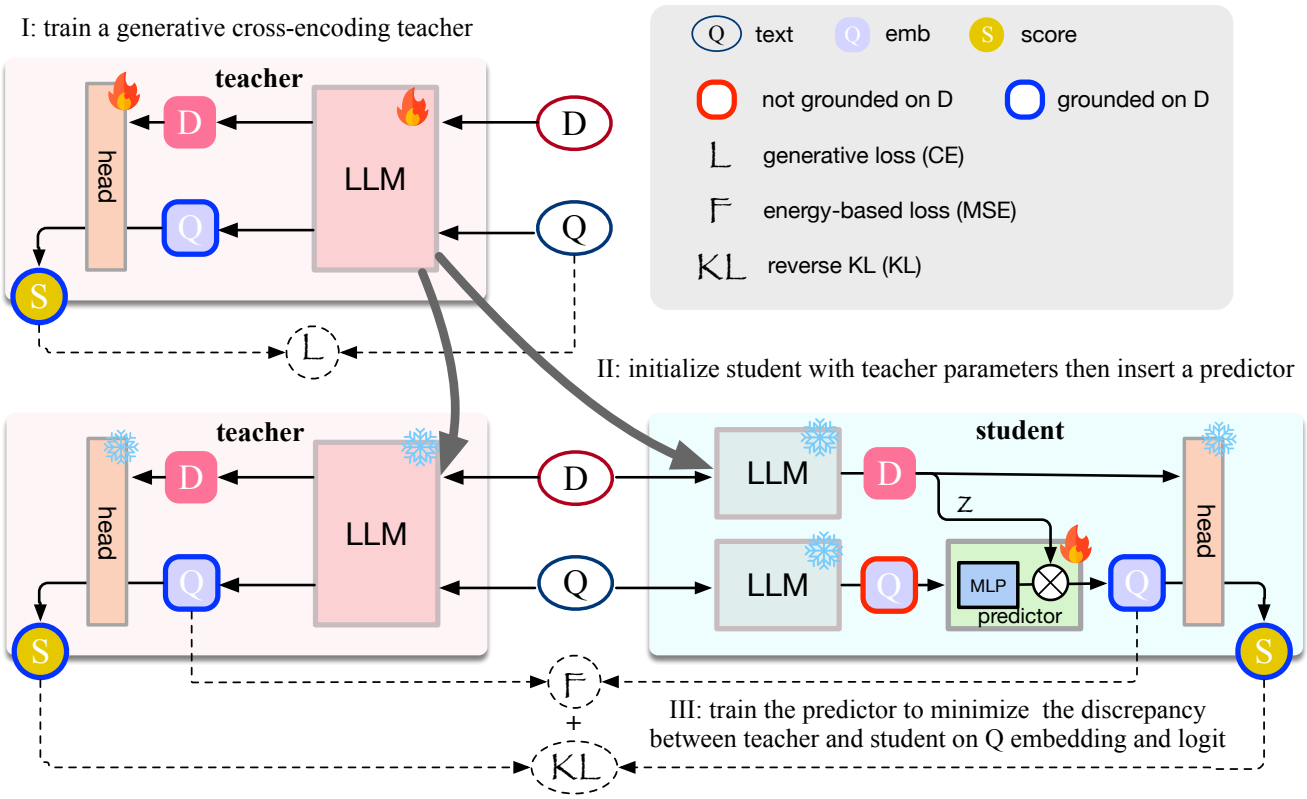


Figure 3: Pipeline to develop Rabtriever. I: A teacher cross-encodes query (Q) and document (D), then generates the retrieval score (s). We train it by finetuning an LLM on question-answering samples using the cross-entropy (CE) loss ($\mathcal{L}$). II: We initialize the student from teacher, with parameters frozen. A predictor is then inserted between LLM layers and heads, consisting of an MLP and a multiplication operator. III: The predictor converts the Q embedding into a new embedding, with the D embedding as the latent control variable (z). The predicted Q embedding is then fed into the head, which produces the relevance score for student. Both embedding and scoring differences between teacher and student are minimized ($\mathcal{F}$).

The Predictor. We implement $\mathcal{P}$ first with a 2-layer MLP with Relu activation, then applying a light-weight elementary multiplication:

$$e_Q^{\text{student}} = \text{MLP}_\theta(e_Q) \odot e_D[-1] \tag{12}$$

where $[-1]$ denotes the last embedding of the sequence. During **student** training, only the MLP parameter θ is trainable. Although all the modules of $\mathcal{P}$ seems simple, later experiments show that they are effective enough for the distillation. Apparently, they are lightweight enough to ensure inference efficiency.

Training. We replace the next-token prediction in the **teacher** with the energy-based modeling (EBM), by minimizing the energy between **teacher** and **student** on the latent space. As suggested by JEPA [2], we implement mean-square error (MSE) instead of a contrastive loss, which helps improve the convergence and stability of EBM. Besides MSE on Q embedding, we augment an auxiliary reverse KL of **student** w.r.t **teacher**, which further pushes the **student** to approximate the **teacher**'s behavior on the logit level:

$$\min_\theta \mathcal{F} := \mathcal{F}^{\text{MSE}}(e_Q^{\text{student}}, e_Q^{\text{teacher}}) + wKL^{\text{rev}}(\text{P}^{\text{student}} || \text{P}^{\text{teacher}})$$

where w is the KL loss weight. Apparently, the energy $\mathcal{F}$ decays to zero when **student** perfectly reconstructs **teacher**. In the ablation study, we verify that both $\mathcal{F}^{\text{MSE}}$ and KL^{rev} are critical for retrieval performance improvement.

During the training, the student is actively updating, rollouts all samples while the teacher is frozen and infers e_Q^{teacher} and $\text{P}^{\text{teacher}}$, which corresponds to the on-policy distillation. The architecture, loss, and training mechanism are visualized in Figure 3 (III).

4.3 Inference Efficiency

Within the predictor, the time complexity of MLP is $\text{O}(L_Q)$ while the dot multiplication is $\text{O}(1)$. The entire time complexity of Rabtriever then becomes

$$\underbrace{n\text{O}(L_D)^2}_{\text{Offline}} + \underbrace{\text{O}(L_Q)^2 + \text{O}(L_Q) + n\text{O}(1)}_{\text{Online}} \tag{13}$$

Although slower than the original bi-encoder method (Eq. (3) has only linear components), apparently, its online time complexity has been much improved from pure cross-encoder methods (Eq. (5)).

Table 2: Performances on three rationale-based retrieval tasks. 'R' denotes Recall. LaHoRe performs better RankLLaMA as reankers; Rabtriever is then distilled from LaHoRe and achieves the best accuracy among retrievers. Bold indicates the best result, while underline indicates the second best. Numbers are in percentages.

	Dataset(→)		ESConv			PsyQA			Saycan				RT ↓
	Method (↓)	Size	R@1	R@3	MRR	R@1	R@3	MRR	R@1	R@3	R@5	MRR	(ms)
Retriever	TART-Contriever [1]	110M	1.4	38.5	35.1	11.3	44.2	36.5	1.2	19.2	36.5	17.5	
	E5-large-instruct [23]	550M	19.2	47.2	40.5	12.9	42.8	36.3	7.7	17.3	26.9	18.2	
	InstructOR XL [19]	1.5B	13.7	44.3	36.4	24.6	61.2	48.1	17.3	26.9	34.6	28.4	
	RepLLaMA [12]	7B	17.4	49.1	39.7	29.6	56.0	49.6	9.6	25.0	38.4	23.6	
	Promptriever [24]	7B	17.6	48.9	39.5	29.6	61.7	50.9	5.8	19.2	34.6	19.0	
	E5-Mistral [23]	7B	15.5	50.2	39.3	11.4	48.0	36.8	14.9	19.1	19.1	17.0	
	GritLM [15]	7B	19.2	50.2	41.7	26.9	57.5	48.4	35.8	43.5	43.1	46.3	
	OneGen [25]	7B	29.0	52.6	41.5	31.3	46.0	46.9	44.2	67.3	75.0	56.8	
	LlaMA2Vec [10]	7B	23.9	55.6	45.5	**34.3**	75.4	**57.5**	17.3	40.4	46.2	31.1	**93**
	Rabtriever (ours)	7B	**32.8**	**66.4**	**54.7**	32.8	**76.1**	55.1	**67.3**	**76.9**	**78.8**	**73.7**	95
Reranker	RankLLaMA [12]	7B	32.3	64.8	52.9	33.6	77.4	57.2	26.9	50.0	75.0	44.9	
	LaHoRe [8] (**teacher**)	7B	**34.4**	**67.1**	**54.8**	**35.6**	**77.6**	**58.2**	**71.2**	**88.5**	**94.2**	**79.8**	980

5 Experiment

5.1 Settings

Implementation. We set the maximum of L_D to 32 and conduct **left-padding** to the document part. The rest of the textual input is **right-padded**, with the total window length of 4096. Rabtriever is initialized from Qwen2-7B-Instruct [17]. Training is conducted using LlamaFactory [28], with 8 A100 GPUs and takes less than 10 hours. The learning rate is $1.0e-6$, the batch size is 16, and the epoch number of epochs The KL loss weight is set to $w = 0.25$. Inference is implemented with vLLM with the prefix cache [9]. For the efficiency analysis, the online response time (RT) is calculated and averaged for each query-document pair in the test set.

Metrics. We evaluate on different retrieval metrics, including Recall@k (k may be set to 1, 3, 5, considering the total number of documents), p-MRR@10, as well as nDCG@10 for ranking-annotated benchmarks such as BEIR. In the following contexts, we use R@k to denote Recall@k, while using MRR to abbreviate p-MRR@10.

Baselines. We compare Rabtriever to typical retrievers based on different architectures or methodologies, including traditional dense retrievers such as *TART-Contriever* [1], *RepLLaMA* [12], *E5-large-instruct*, and *E5-Mistral* [23]; instructional retrievers such as *InstructOR XL* [19], *Promptriever* [24] and *LLaMA2Vec* [10]; and multi-task retrievers like *GritLM* [15] and *OneGen* [25], which integrates the representative and generative tasks. For rerankers, we experiment with *RankLLaMA*, the cross-encoder variant of *RepLLaMA* [12]; as well as *LaHoRe* [8], which behaves as the **teacher** of distillation in this paper. For ease of comparison, most baselines have an identical parameter size (7B) to our implementation, as highlighted in Table 2.

5.2 Rationale-based Retrieval

Datasets. In this subsection, we conduct the experiments on two typical scenarios of rational-based retrieval:

(1) Empathetic conversation. These datasets focus on emotional support and psychological counseling, where the system responses are annotated with predefined, optimal support strategies.

- **ESConv** [11] is a multi-turn emotional support dialogue dataset containing 1.2k conversations and 12.7k utterances. Each supporter response is annotated with one of eight fine-grained support strategies, enabling the study of long-horizon strategy planning in empathetic conversations.
- **PsyQA** [20] is a single-turn psychological counseling dataset consisting of 4,012 question–answer pairs. Each response is labeled with one of seven counseling strategies, emphasizing the strategy selection under limited conversational contexts.

For both datasets, the query is formalized by prepending the historical context (if any) to each user utterance, while all strategy candidates form the pool of documents.

(2) Robotic manipulation. In this scenario, the robot makes decisions via language grounding, each time choosing the best subskill from all textual skill candidates, in order to achieve its long-term goal.

- **SayCan** [7] is a large-scale robotic manipulation dataset with approximately 276K trajectories, each consisting of up to 50 steps and covering 551 low-level skills. The dataset supports research on grounding natural language instructions into executable action sequences through hierarchical planning.

Retrieval Results. Table 2 shows retrieval performances on three rationale-based datasets. Although fast, most of retriever baselines fail to capture cross-semantic relations between queries and documents, with generally lower accuracies. Although this issue can be alleviated by integrating a generative subtask, such as GritLM and OneGen; or prompting the LLM for next-sentence inference, as did by LlaMA2Vec, they still perform worse than LaHoRe, the cross-encoding **teacher**. In contrast, our **Rabtriever** improves the performance significantly, revealing the effectiveness of distillation.

Compared to the teacher and bi-encoders. An important research question can be further proposed: **if Rabtriever effectively trades off the accuracy of reranker and the computational efficiency**

Table 3: Ablation studies on PsyQA and SayCan.

Dataset(→)	PsyQA			SayCan			
Method (↓)	R@1	R@3	MRR	R@1	R@3	R@5	MRR
w/o-mul	2.0	10.7	21.3	9.6	15.4	19.2	14.7
w/o-$\mathcal{F}^{\text{MSE}}$	0.0	0.7	17.2	23.1	40.4	44.2	33.1
w/o-KL^{rev}	24.1	24.9	37.3	21.2	30.8	46.2	31.8
w/-KL^{forward}	28.4	55.3	48.5	46.8	54.1	62.4	59.7
Rabtriever	**32.8**	**76.1**	**55.1**	**67.3**	**76.9**	**78.8**	**73.7**

of bi-encoder retrievers? From Table 2, one can observe that Rabtriever's performance is relatively close to **teacher**, indicating the knowledge is effectively preserved during the distillation. For Saycan, Rabtriever even apparently outperforms another reranker, RankLLaMA. Table 2 also compares the average response time (RT) of Rabtriever, **teacher**, and the best retriever baseline, LlaMA2Vec, in milliseconds. Rabtriever has a similar RT to LlaMA2Vec, while is much faster than **teacher**, which validates its retrieval efficiency.

Ablation Study. To verify the necessity of method components, we first attempt to remove the MSE loss (*w/o-*$\mathcal{F}^{MSE}$), the KL divergence (*w/o-*KL^{rev}), and the multiplication operator (*w/o-mul*). We also replace the reverse KL with forward KL (*w/-*$KL^{forward}$), to validate the benefit of reverse KL within the on-policy distillation.

Table 3 indicates that Rabtriever outperforms all these ablations, validating the reasonability of our framework. Especially, *w/o-mul* completely fails, because the multiplication operation introduces the grounding information of D on Q, which is critical to reconstruct the cross-encoding distribution. Removing the reverse KL term, or switching it to the forward KL, resulting in significant performance degradation, verifying that reverse KL significantly helps the **student** focus on the **teacher**'s critical behaviors (answer True or False solely), facilitating the convergence by conditioning on **student**'s own distribution. Finally, removing the latent difference loss ($\mathcal{F}^{\text{MSE}}$) resulting in varied impacts, where the PsyQA performance is much more affected than SayCan. We suppose the reason is that PsyQA infers much longer context window (with the conversation history) than SayCan, in which situations distillation on the latent space becomes more critical for model convergence.

The RT dependency. As derived in Eq. (13), our theoretical efficiency analysis indicates that Rabtriever achieves a *linear* dependency on the document length (L_D), in contrast with the *quadratic* dependency of cross-encoder **teacher** (Eq. (5)). To provide the empirical verifications, we provide the statistics of online RTs with respect to different values of L_D. In Figure 4, one can clearly observe the linear dependency of **teacher** and the quadratic dependency of **teacher** on L_D, verifying the theoretical conclusion.

Scalability. Figure 5 exhibits the SayCan performance of Rabtriever on smaller backbones, including 0.5B and 1.5B. Compared to LlaMA2Vec which has a 7B backbone, Rabtriever performs similarly, even with much smaller backbones. This promising result not only verifies the scalability of Rabtriever, but also reveals the possibility of further improving the retrieval speed at a tradeoff with performance, by using smaller backbones if necessary.

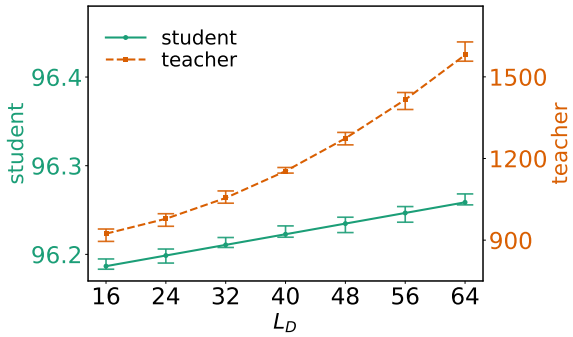


Figure 4: Averaged Online RTs corresponding to different L_D.

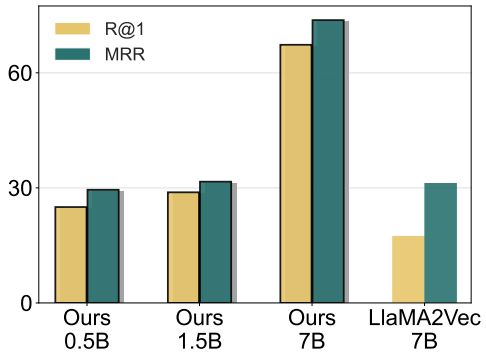


Figure 5: Method scalability on SayCan.

5.3 Generalize to Fact-based Retrieval

We finally verify the generalization capability of Rabtriever on two fact-based retrieval benchmarks, comparing with SoTA baselines.

Settings. For in-domain test, we evaluate on MS MARCO [16] passage retrieval tasks, which consists of approximately 500k training examples. MRR@10 are reported on the *dev* split. For out-of-domain test, models trained by MS MARCO are evaluated on BEIR [22], in a zero-shot manner. We test on the 13 subtasks of BEIR which are publicly available, and report the nDCG@10 results.

In-domain performance. Figure 6 compares the MS MARCO results of different methods. The **teacher** outperforms all baselines with MRR of 52.4, indicating stronger comprehensions from cross-encoding rerankers. Furthermore, our **student** also obtains an MRR of 49.30, surpassing all retriever baselines (although less than **teacher**). This promising result shows that our distillation methodology can also effectively adapt to generalized retrieval tasks, while improve the inference speed of cross-encoder dramatically.

Zero-shot results. We finally exhibit the zero-shot results on BEIR in Table 4. Again, the **teacher** obtained the best averaged nDCG@10 across all retrievers, performing the best on 5/13 subtasks. On the other hand, the **student** obtains a comparable performance to the cross-encoder rerankers, and has the second-best averaged result among all bi-encoder retrievers. Especially, the **student** performs

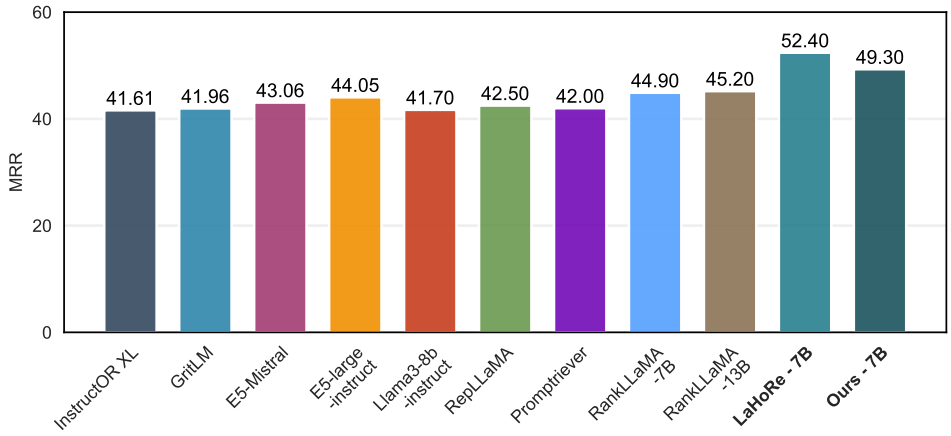


Figure 6: MRR comparison of models on MS MARCO passage retrieval.

Table 4: Out of domain performance on BEIR (nDCG@10). Best value in the row is bolded. The best result is marked bolded and the second best is underlined. Compared to the best reranker LaHoRe, Rabtriever obtains averagely 58.2, and performs the best in 2/13 subtasks and second-best in 4/13 subtasks, which is close to the best retriever competitor (GritLM).

	Dataset	Retriever							Reranker		**Ours**
		BM25 N/A	E5-large 550M	E5-Mistral 7B	Promptriever 7B	RepLLaMA 7B	LlaMA2Vec 7B	GritLM 7B	RankLLaMA 7B	LaHoRe 7B	Rabtriever 7B
Dataset has dev/train set	DBPedia	29.9	41.3	48.9	45.2	44.8	45.9	46.6	48.3	**75.5**	62.7
	FEVER	48.1	82.8	**87.8**	82.8	82.9	81.3	82.7	83.9	84.9	76.7
	FiQA	25.1	43.8	56.6	46.6	45.0	48.5	60.0	46.5	**91.4**	54.9
	HotpotQA	56.9	71.2	75.7	69.5	68.8	70.1	79.4	75.3	76.9	**80.0**
	NFCorpus	32.1	34.0	38.6	36.9	36.0	38.2	**41.2**	30.3	40.6	40.6
	Quora	80.4	88.2	**89.6**	88.0	86.0	88.3	89.5	85.0	88.2	88.5
	SciFact	68.7	70.4	76.4	76.3	75.3	74.8	**79.2**	73.2	78.9	78.9
No dev/train set	Arguana	36.6	54.4	61.9	56.7	48.6	56.5	**63.2**	56.0	36.8	36.8
	Climate-FEVER	13.6	25.7	38.4	32.1	29.3	38.2	30.9	28.0	**42.3**	23.7
	NQ	28.5	64.1	63.5	62.6	63.0	64.6	70.3	66.3	**92.0**	73.1
	SCIDOCS	15.8	17.5	16.3	19.7	16.1	18.9	24.4	17.8	17.1	**25.0**
	TREC-COVID	62.3	71.3	87.2	84.6	83.9	**86.9**	74.8	85.2	81.6	81.8
	Touche-2020	33.1	23.4	26.4	32.0	34.1	34.2	27.9	40.1	**49.9**	34.3
	# of best	0	0	2	0	0	1	3	0	5	2
	# of 2nd-best	0	0	2	0	0	0	4	2	3	4
	Average	40.9	53.9	59.0	56.4	54.9	57.4	59.2	56.6	**65.8**	58.2

the best on 2/13 subtasks and the second best on 4/13 subtasks, which is close to GritLM, the most competitive bi-encoder baseline.

6 Conclusion

In this paper, we propose an efficient, LLM-based retriever called Rabtriever, to address the challenging problem of rationale-based retrieval. We balance the retrieval accuracy and the computational efficiency by implementing a bi-encoder architecture, while providing a retrieval score that considers the cross-semantic information between query and document. We achieve this target by on-policy distilling from a cross-encoder reranker, based on the JEPA framework, with both latent MSE and reverse KL losses implemented. Experiments verify that our Rabtriever can handle rationale-based retrieval problems, surpassing bi-encoder retriever baselines. Compared to cross-encoding rerankers, our Rabtriever performs similarly while reducing the *quadratic* time complexity on the document length to *linear*. We further generalize Rabtriever to traditional fact-based retrieval tasks, and also observe state-of-the-art performances obtained on both MS MARCO and BEIR benchmarks.

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